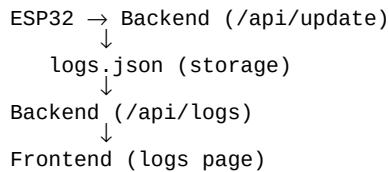


Phase 3: Database & Logging (Detailed Plan with Code)

System Flow Diagram:



Member 1: Database Setup

Create storage file to store logs permanently.

Create folder:
software/backend/data/

Create file:
logs.json

Add:
[]

Member 2: Backend API

Modify server.js to store and fetch logs.

```
const fs = require("fs");
const path = "../data/logs.json";

// Read logs
function readLogs() {
  return JSON.parse(fs.readFileSync(path));
}

// Write logs
function writeLogs(data) {
  fs.writeFileSync(path, JSON.stringify(data, null, 2));
}

// Update API
app.post("/api/update", (req, res) => {
  const logs = readLogs();

  logs.unshift({
    time: new Date().toLocaleString(),
    status: req.body.status
  });

  writeLogs(logs);

  res.json({ success: true });
});

// Get logs
app.get("/api/logs", (req, res) => {
  res.json(readLogs());
});
```

Member 3: Frontend Logs Page

Fetch logs from backend and display.

```
async function loadLogs() {
  const res = await fetch("http://localhost:5000/api/logs");
  const data = await res.json();

  const table = document.getElementById("logTable");
  table.innerHTML = "";

  data.forEach(log => {
```

```
        table.innerHTML += `
        <tr>
          <td>${log.time}</td>
          <td>${log.status}</td>
        </tr>`;
      });
    }

    setInterval(loadLogs, 3000);
    loadLogs();
```

Member 4: Integration & Testing

Ensure everything works together.

1. Run backend:
node server.js
2. Open frontend
3. Test flow:
Click simulate → logs update → refresh page → logs remain
4. Check:
No console errors

Final Goal

Logs are stored permanently and displayed correctly in UI.